University of Michigan
Introduction to Data Science in Python

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Introduction to Data Science in Python

by University of Michigan

About this Course

This course will introduce the learner to the basics of the python programming environment, including fundamental python programming techniques such as lambdas, reading and manipulating csv files, and the numpy library. The course will introduce data manipulation and cleaning techniques using the popular python pandas data science library and introduce the abstraction of the Series and DataFrame as the central data structures for data analysis, along with tutorials on how to use functions such as groupby, merge, and pivot tables effectively. By the end of this course, students will be able to take tabular data, clean it, manipulate it, and run basic inferential statistical analyses. This course should be taken before any of the other Applied Data Science with Python courses: Applied Plotting, Charting & Data Representation in Python, Applied Machine Learning in Python, Applied Text Mining in Python, Applied Social Network Analysis in Python.

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Taught by:

Christopher Brooks
Associate Professor, School of Information

Basic Info	Course 1 of 5 in the Applied Data Science with Python Specialization
Level	Intermediate
Language	English Subtitles: Arabic, French, Bengali, Ukrainian, Chinese (Simplified), Greek, Italian, Portuguese (Brazil), Vietnamese, Dutch, Korean, German, Pashto, Urdu, Russian, Thai, Indonesian, Swedish, Turkish, Azerbaijani, Spanish, Dari, Hindi, Japanese, Kazakh, Hungarian, Polish
How To Pass	Pass all graded assignments to complete the course.
User Ratings	★ 4.5

Syllabus

Module 1

Fundamentals of Data Manipulation with Python

In this module, you'll get an introduction to the field of data science, review common Python functionality and features that data scientists use, and be introduced to the Coursera Jupyter Notebook for the lectures. All of the course information on grading, prerequisites, and expectations are on the course syllabus, and you can find more information about the Jupyter Notebooks on our Course Resources page.

20 videos, 4 readings

1. Video: Introduction to Specialization
2. Video: Introduction to the Course
3. Reading: Syllabus
4. Reading: Pre-Course Survey
5. Video: The Coursera Jupyter Notebook System
6. Ungraded Lab: Your Personal Jupyter Notebook Workspace
7. Video: Python Functions
8. Video: Python Types and Sequences
9. Video: Python More on Strings
10. Video: Python Demonstration: Reading and Writing CSV files
11. Video: Python Dates and Times
12. Video: Advanced Python Objects, map()
13. Video: Advanced Python Lambda and List Comprehensions
14. Graded Assignment: Practice Quiz: Python Programming
15. Ungraded Lab: Module 1 Jupyter Notebooks
16. Video: Creating Array
17. Video: Manipulating Array
18. Video: From Arrays to Images and Back
19. Video: Indexing, Slicing, and Iterating
20. Video: Trying NumPy with Datasets
21. Graded Assignment: Practice Quiz: Numerical Python Library (NumPy)
22. Video: Regex Matching and Anchors
23. Video: Patterns and Character Classes
24. Video: Quantifiers
25. Video: Groups
26. Video: Advanced Assertions and Applications
27. Reading: Regular Expression Operations Documentation
28. Ungraded Plugin: Regex Practice Session
29. Reading: Module 1 Textbook Reading Assignment

Graded: Quiz 1
Graded: Assignment 1

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Module 2

Basic Data Processing with Pandas

In this module of the course, you'll learn the fundamentals of one of the most important toolkits Python has for data cleaning and processing -- pandas. You'll learn how to read in data into DataFrame structures, how to query these structures, and the details about such structures are indexed.

13 videos, 1 reading

1. Ungraded Lab: Module 2 Jupyter Notebooks
2. Video: Introduction to Pandas
3. Video: The Series Data Structure
4. Video: Creating Pandas' Series
5. Video: Querying a Series
6. Video: Vectorized Operations
7. Video: Appending Series
8. Graded Assignment: Practice Quiz: Pandas and Series Data
9. Video: DataFrame Data Structure
10. Video: DataFrame Indexing and Loading
11. Video: Querying a DataFrame
12. Video: Indexing Dataframes
13. Video: Missing Values, Part 1
14. Video: Missing Values, Part 2
15. Video: Example: Manipulating DataFrame
16. Graded Assignment: Practice Quiz: DataFrame
17. Reading: Module 2 Textbook Reading Assignment

Graded: Quiz 2
Graded: Assignment 2

Module 3

More Data Processing with Pandas

In this module, you'll deepen your understanding of the python pandas library by learning how to merge DataFrames, generate summary tables, group data into logical pieces, and manipulate dates. We'll also refresh your understanding of scales of data, and discuss issues with creating metrics for analysis. The week ends with a more significant programming assignment.

12 videos, 1 reading

1. Ungraded Lab: Module 3 Jupyter Notebooks
2. Video: Merging Dataframes
3. Video: Handling Conflicts between Dataframes
4. Video: Concatenating DataFrames
5. Video: Pandas Idioms, Part 1
6. Video: Pandas Idioms, Part 2
7. Video: Group by, Part 1
8. Video: Group by, Part 2
9. Video: Group by, Part 3
10. Video: Scales
11. Video: Pivot Table
12. Video: Date/Time Functionality
13. Video: Working with Dates in a Dataframe
14. Graded Assignment: Practice Quiz: More Data Processing with Pandas
15. Reading: Module 3 Textbook Reading Assignment

Graded: Quiz 3
Graded: Assignment 3

Module 4

Answering Questions with Messy Data

In this final module of the course, you'll be introduced to a variety of statistical techniques such a distributions, sampling and t-tests. The week ends with two discussions of science and the rise of the fourth paradigm -- data driven discovery.

2 videos, 6 readings

1. Ungraded Lab: Module 4 Jupyter Notebooks
2. Video: Basic Statistical Testing
3. Video: Other Forms of Structured Data
4. Reading: Science Isn't Broken: p-hacking
5. Reading: Goodhart's Law
6. Reading: The 5 Graph Algorithms that you should know
7. Reading: Post-Course Survey
8. Reading: Keep Learning with Michigan Online!
9. Reading: Progress Note from Christopher Brooks

Graded: Assignment 4
Graded: Final Quiz

How It Works

General

What do start dates and end dates mean?
Once you enroll, you'll have access to all videos, readings, quizzes, and programming assignments (if applicable). If you choose to explore the content without purchasing, you may not be able to access certain assignments. If you don't finish all graded assignments before the end of the session, you can reset your deadlines. Your progress will be saved and you'll be able to pick up where you left off.

What are due dates? Is there a penalty for submitting my work after a due date?
Within a course, there are suggested due dates to help you manage your schedule and keep work from piling up. Quizzes and programming assignments can be submitted late without consequence. However, it is possible that you won't receive a grade if you submit your peer-graded assignment too late because classmates usually review assignment within three days of the assignment deadline.

Can I re-attempt an assignment?
Yes. If you want to improve your grade, you can always try again. If you're re-attempting a peer-graded assignment, re-submit your work as soon as you can to make sure there's enough time for your classmates to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review learning material during this delay.

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Programming assignments

Programming assignments require you to write and run a computer program to solve a problem.

What are programming assignments?
Programming assignments include both assignment instructions and assignment parts. Instructions may include a link to a downloadable starter package that includes starter code, detailed guidelines, and other resources. Assignment parts are similar to individual quiz questions. Each part is a single coding task that can be completed one at a time.

How are programming assignments graded?
Programming assignments are graded automatically. If they use a built-in-algorithm you'll see your grade within seconds. If they use a custom grader, you may need to wait up to an hour.

Can I resubmit a programming assignment?
You can resubmit all programming assignments to improve your grade. Follow the same steps as submitting a new assignment.

What do I do if I have trouble submitting my assignment?
If you have trouble submitting your assignment, we encourage you to visit the Discussion Forums as many of your peers are likely to have had similar problems and have found a solution. Each programming assignment has its own sub-forum to discuss with peers.

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Course 1 of Specialization

Gain new insights into your data

Learn to apply data science methods and techniques, and acquire analysis skills.

University of Michigan
Applied Data Science with Python

View the course in catalog

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